EXPERIMENT 4: GENERATE CONTEXTUAL EMBEDDINGS USING A PRE-TRAINED TRANSFORMER

Aim

To generate contextual embeddings for a given sentence using a pre-trained BERT Transformer model.

Algorithm

1. Import PyTorch, AutoTokenizer, and AutoModel.
2. Load the pre-trained BERT tokenizer and model.
3. Provide an input sentence.
4. Convert the sentence into token IDs.
5. Pass the tokenized input to the BERT model.
6. Extract the last hidden state.
7. Display the embedding shape and the embedding vector of the first token.

Program

```
# Experiment 4: Generate Contextual Embeddings
import torch
from transformers import AutoTokenizer, AutoModel
# Load tokenizer and model
tokenizer = AutoTokenizer.from_pretrained(
    "bert-base-uncased"
)
model = AutoModel.from_pretrained(
    "bert-base-uncased"
)
# Input sentence
text = "Artificial intelligence is transforming education."
# Tokenize the input
inputs = tokenizer(
    text,
    return_tensors="pt"
)
# Generate contextual embeddings
model.eval()
```

```
with torch.no_grad():
    outputs = model(**inputs)

# Extract last hidden state
embeddings = outputs.last_hidden_state

# Display results
print("Input Text:", text)

print("\nTokenized Input:")
print(tokenizer.convert_ids_to_tokens(inputs["input_ids"][0]))

print("\nEmbedding Shape:")
print(embeddings.shape)

print("\nFirst Token Embedding:")
print(embeddings[0, 0, :])

print("\nContextual Embeddings Generated Successfully.")
```

Input

Artificial intelligence is transforming education.

Expected Output

```
Input Text: Artificial intelligence is transforming education.

Tokenized Input:
['[CLS]', 'artificial', 'intelligence', 'is',
 'transforming', 'education', '.', '[SEP]']

Embedding Shape:
torch.Size([1, 8, 768])

First Token Embedding:
tensor([...])
```

Contextual Embeddings Generated Successfully.

The embedding vector is abbreviated here because it contains 768 numerical values. Your program will display the actual values.

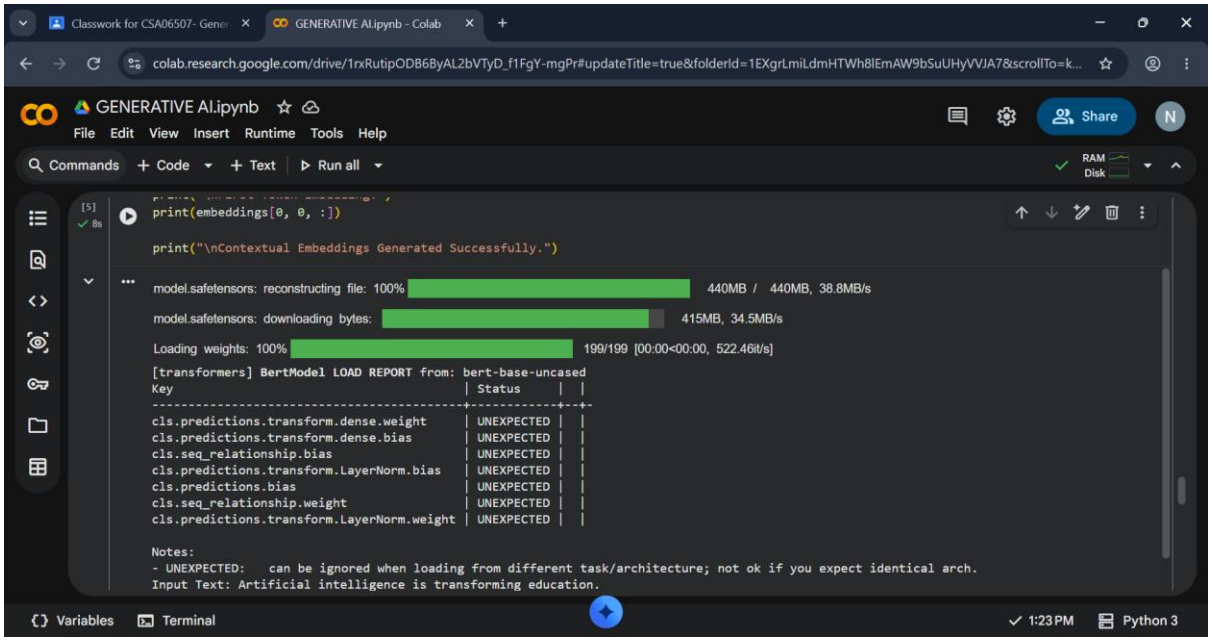
Explanation

The embedding shape is:

[1, 8, 768]

- 1 — Number of input sentences.
- 8 — Number of tokens, including special tokens.
- 768 — Dimensions of each token's embedding.

A **contextual embedding** represents a token based on the surrounding text. For example, a word's representation can change depending on how it is used in a sentence.



Result

Thus, contextual embeddings were generated successfully using a pre-trained BERT model, and the dimensions of the resulting embeddings were examined.